MATH INQUIRERS HOLIDAY PROGRAM

Readiness Tracker and Screening Set

Spot children ready for olympiad-style work: four signs to track and an 8-question screen

Notice it. Test it. Explain why.

Who is ready to go further? We look at how a child thinks, not only the score. Tick after each lesson (30 seconds per child).

The four signs

Code	Sign	What you see or hear
A	Solved a Challenge question alone	Got the answer without hints, and showed how (counters, drawing or words).
B	Found a second way	Used a different method or found another answer, and said how it is different.
C	Noticed a pattern and said why	Said 'it always goes up by 3 because...', not just 'it goes up'.
D	Stretched it to a new case	Said what would happen with a bigger number or a changed rule, and checked it.

Lesson log: copy one page per lesson

Lesson number: _____ Date: _____ Tutor: _____

Child	A	B	C	D	Total	Note (what they said or did)
1.						
2.						
3.						
4.						
5.						
6.						
7.						

8.						

Programme summary: one row per child

Write each lesson's total (0 to 4) in the box. Then add up in the last columns.

Child	L1	L2	L3	L4	L5	L6	L7	L8	L9	L10	Lessons with 3+	Decision
1.												
2.												
3.												
4.												
5.												
6.												
7.												
8.												

How to decide

- 3 or 4 in at least 3 lessons: invite to the Challenge Group and give the Screening Set below.
- 2 in most lessons: keep going and give more Think Harder questions in class.
- Mostly 0 or 1: do not worry. Check understanding of the core questions first. Many children grow later in the programme.
- One lesson alone means little. Look at the pattern across lessons, and at what the child actually says.
- Use this to guide extra challenge, not to label a child. Do not share rankings between families.

Challenge Screening Set: child sheet

Name: _____ Date: _____

Take your time. For each question, show how you found your answer with a drawing, counters or words. Then tell your tutor why.

1. Frog A jumps in 3s from 0. Frog B jumps in 4s from 0. Which numbers up to 40 do both frogs land on? (Do not count 0.)

_____

2. 12 children voted for three fruits. No two fruits got the same number of votes. The most votes was 5. How many votes did each fruit get?

_____

3. Tiles: the 1st pattern has 3 tiles, the 2nd has 5, the 3rd has 7. How many tiles in the 10th pattern? Which pattern has 31 tiles?

_____

4. I think of a number. I add 6, take away 4, then add 5. I get 17. What is my number?

_____

5. You have one 5¢, one 10¢ and one 20¢ coin. How many different totals can you make using one, two or all three coins? List them.

_____

6. A big triangle has 1 line from the top corner to the bottom. How many triangles? What about 2 lines? 3 lines?

_____

7. Four friends stand in a line. Ali is not first. Ben is ahead of Ali. Sam is last. Mei is behind Ali. In what order do they stand?

_____

8. How many ways can you make 30¢ using 20¢, 10¢ and 5¢ coins?

_____

Screening Set: tutor guide and answer key

- Use one to one or in a group of up to 4. Allow about 25 minutes. Read each question aloud if needed.
- Do not give hints in the first attempt. After 2 minutes you may give one hint, and score it as 'with a hint'.
- Score each question: 2 = correct and explains why (without hint); 1 = correct, or right idea with a hint; 0 = not yet.
- Then ask the extension question once. A good answer earns a star (★). Stars show how far the child can stretch.
- Total out of 16. As a guide, 12 or more with 3 or more stars suggests a child who would enjoy olympiad-style problems. Combine this with the lesson tracker; neither alone decides.
- These problems use the same kinds of thinking that later maths olympiad papers reward: patterns, listing in order, working backwards and logic. This set is not an official test and is not linked to any school or MOE programme.

#	Question	Answer	Extension (★)
1	Frog A jumps in 3s from 0. Frog B jumps in 4s from 0. Which numbers up to 40 do both frogs land on? (Do not count 0.)	12, 24 and 36.	What if the frogs jumped in 3s and 4s up to 60? How do you know before you count?

2	12 children voted for three fruits. No two fruits got the same number of votes. The most votes was 5. How many votes did each fruit get?	5, 4 and 3.	What if there were 15 children and the most votes was 6? Can the votes still all be different?
3	Tiles: the 1st pattern has 3 tiles, the 2nd has 5, the 3rd has 7. How many tiles in the 10th pattern? Which pattern has 31 tiles?	21 tiles in the 10th. 31 tiles is the 15th.	Which pattern would have 61 tiles? How do you know?
4	I think of a number. I add 6, take away 4, then add 5. I get 17. What is my number?	10 (17 − 5 + 4 − 6)	Make up your own puzzle like this and give it to your tutor.
5	You have one 5¢, one 10¢ and one 20¢ coin. How many different totals can you make using one, two or all three coins? List them.	7 totals: 5, 10, 15, 20, 25, 30, 35.	What if you also had a 50¢ coin? How many totals now?
6	A big triangle has 1 line from the top corner to the bottom. How many triangles? What about 2 lines? 3 lines?	3, 6 and 10 triangles.	How many triangles with 4 lines? What do you think the pattern is?
7	Four friends stand in a line. Ali is not first. Ben is ahead of Ali. Sam is last. Mei is behind Ali. In what order do they stand?	Ben, Ali, Mei, Sam.	Make up another clue so that only one order is possible.
8	How many ways can you make 30¢ using 20¢, 10¢ and 5¢ coins?	6 ways.	What if the coins were 20¢, 10¢ and 5¢ and the total was 40¢?